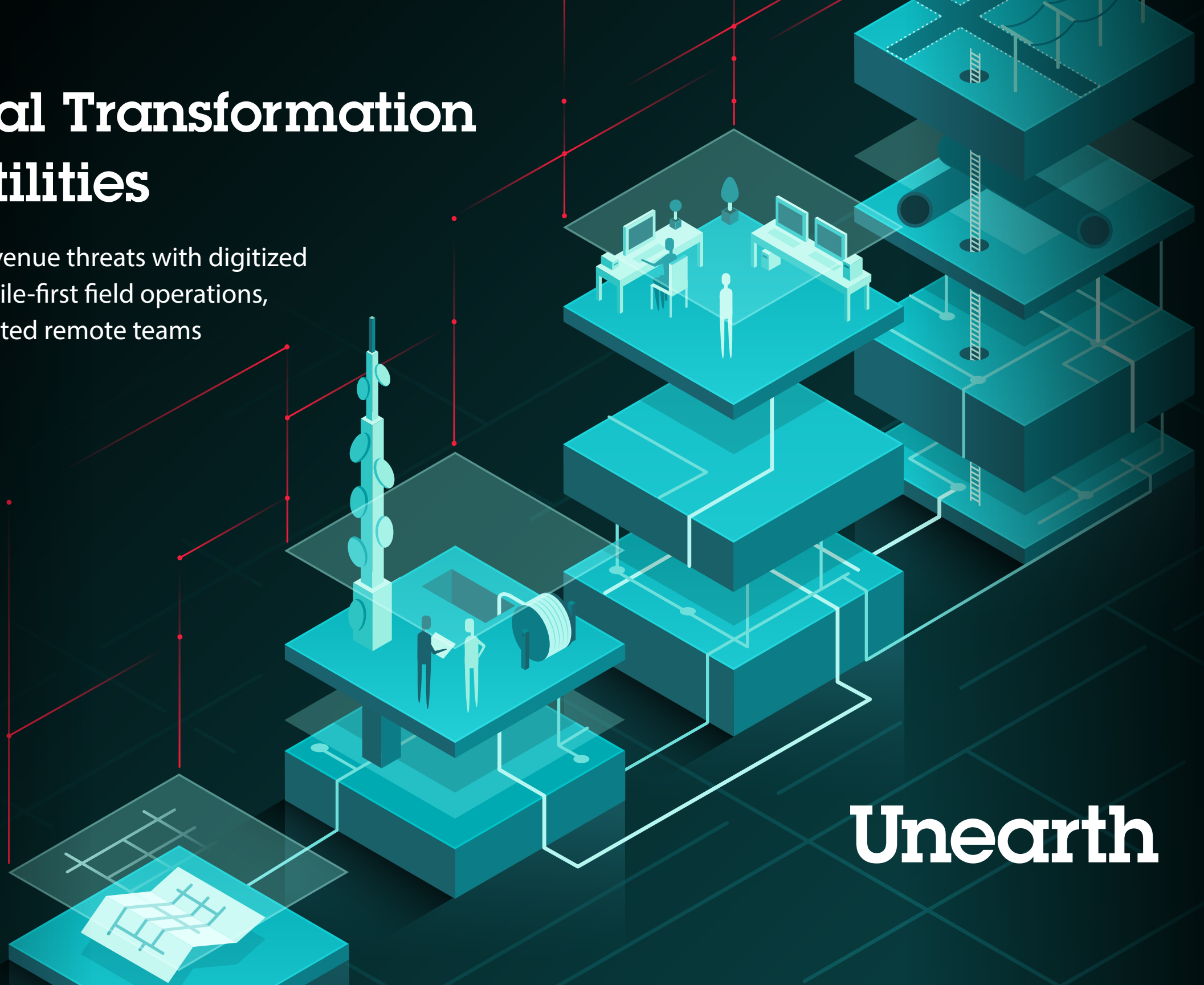


Digital Transformation for Utilities

Mitigate revenue threats with digitized assets, mobile-first field operations, and connected remote teams



Unearth



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Why Digital Transformation is Important for Utilities

Being proactive about digital transformation leads to increased revenue and profit. The opposite is true for those who lag behind.

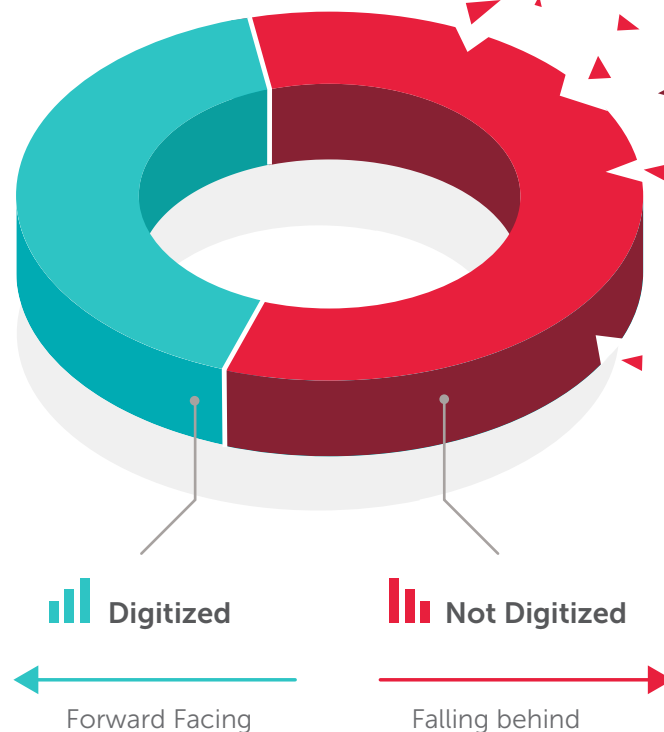
When looking at industries like media, retail, and tech, it's easy to assume that digital technology is deployed evenly throughout our economy. However, those industries are only a small slice of the pie, and when it comes to digital transformation - they are actually the minority.

Industries as a whole are less than **40% digitized**. So, despite the near ubiquity of digital technology in some sectors, the majority have fallen behind.

Utilities are among these digital stragglers.

This is problematic. In the race towards efficiency, there are clear winners and losers - and the winners collect considerable, disproportionate gains.

While utilities are behind in their digital transformation, they're far from back of the pack. The potential for both losses and gains is huge and there's still time to close the digital gap.¹



Digital Transformation Explained

Digital transformation is the process by which organizations update or even re-imagine their operational approach.

Effective digital transformation is all encompassing: spanning day-to-day processes, customer interactions, and internal culture.

One common misconception is that digital transformation is about making the same process faster. Though that can be an effect, it's not necessarily the goal.

Digital transformation is a perspective, a lens through which to view every aspect of a business.

Yes, organizations can use technology to speed up an existing process. However, the larger question should be: Can digital technology rewrite or even eliminate that process completely?

II Utility Digital Transformation: Current Outlook, Roadblocks, and Threats

Current state of digital transformation in utilities

Utilities have achieved a “moderate” level of digitization, but are still behind many other industries.²

Though ahead on business processes and usage transactions, utilities have more work to do when it comes to digitizing customer interaction and asset management.

Physical assets are where utilities have the most room for digital growth.³

At a macro level, industries tend to focus 60% of digital investments on products and services. In contrast, only 43% is funneled towards supply chain modernization.⁴

Utilities follow this same trend. This is especially unfortunate because supply chain physical assets are at the core of utility operations.

All this said, when it comes to digital transformation, utilities do face significant challenges.

Roadblocks to digital transformation

Dispersed physical assets are the primary roadblock to digital transformation in utilities.⁵

Utility assets are large, complex, and geographically widespread. They’re also incredibly expensive. Minimizing operational risk to maximize asset lifespan is the frame by which utilities make decisions.

Analog mindset, a byproduct of this focus on physical assets, presents another challenge. Utility assets have (and always will) exist in the non-digital world, so there’s significant historical precedent for an analog operational approach.

This makes it difficult to execute digital initiatives and attract digitally fluent employees. Processes have looked the same for a long time, so it can be difficult to convince leadership and employees to consider a new approach.

Supply Chain
Modernization

Digital
Investment

Threats to revenue, profit, and growth

Threats like stagnant revenue, significant capital expenditures, and high technology expectations from consumers provide even greater incentive for utilities to pursue digital transformation.

Each of these threats can be mitigated and addressed with digital technology. Utilities that embrace the process fully will see significant gains. The alternative is to stagnate and watch as profits decline.



Stagnant or decreasing revenue

Utility revenue once had a positive relationship with gross domestic product (GDP). As GDP went up, so did revenue. However, advances in energy efficient technology have disrupted this trend - causing revenues to stagnate and profits to decrease.



Significant capital expenditures

Utility asset networks require consistent maintenance. Moreover, as the climate becomes increasingly unpredictable, utilities must make every effort to fortify weaker assets. When compared to other industries, these factors combine to create higher than average capital expenditures.⁶



Higher technology expectations from consumers

As consumers become tech literate, they begin to expect more from service providers. Utility technology startups have disrupted markets and experienced significant positive feedback by using digital technology to streamline consumer interactions.

Benefits of Digital Transformation⁷

- 1 Increase productivity and efficiency**
Using smart meters, cloud-based mapping, and improved data collection tools - utilities can, eventually, create highly efficient smart grids.
- 2 Increase profits and/or lower revenue requirements**
Increases in operational efficiency will go far in addressing revenue stagnation.
- 3 Increase market share**
Organizations that fully commit to digital transformation will experience disproportionate gains. As lagging competitors fall behind, opportunities will arise for future-forward organizations to take the lead.

III

3 Golden Rules for Successful Digital Transformations

1

Align digital strategy with business objectives

To be effective, digital strategy must be tied to business objectives. Without the framework of measurable goals, digital transformation can prove unsustainable.

Getting an initiative off the ground offers the rush and excitement of trying something new. However, as time passes, employees and leaders move on, and that initial rush fades. It's all too easy for commitment to fade as well.

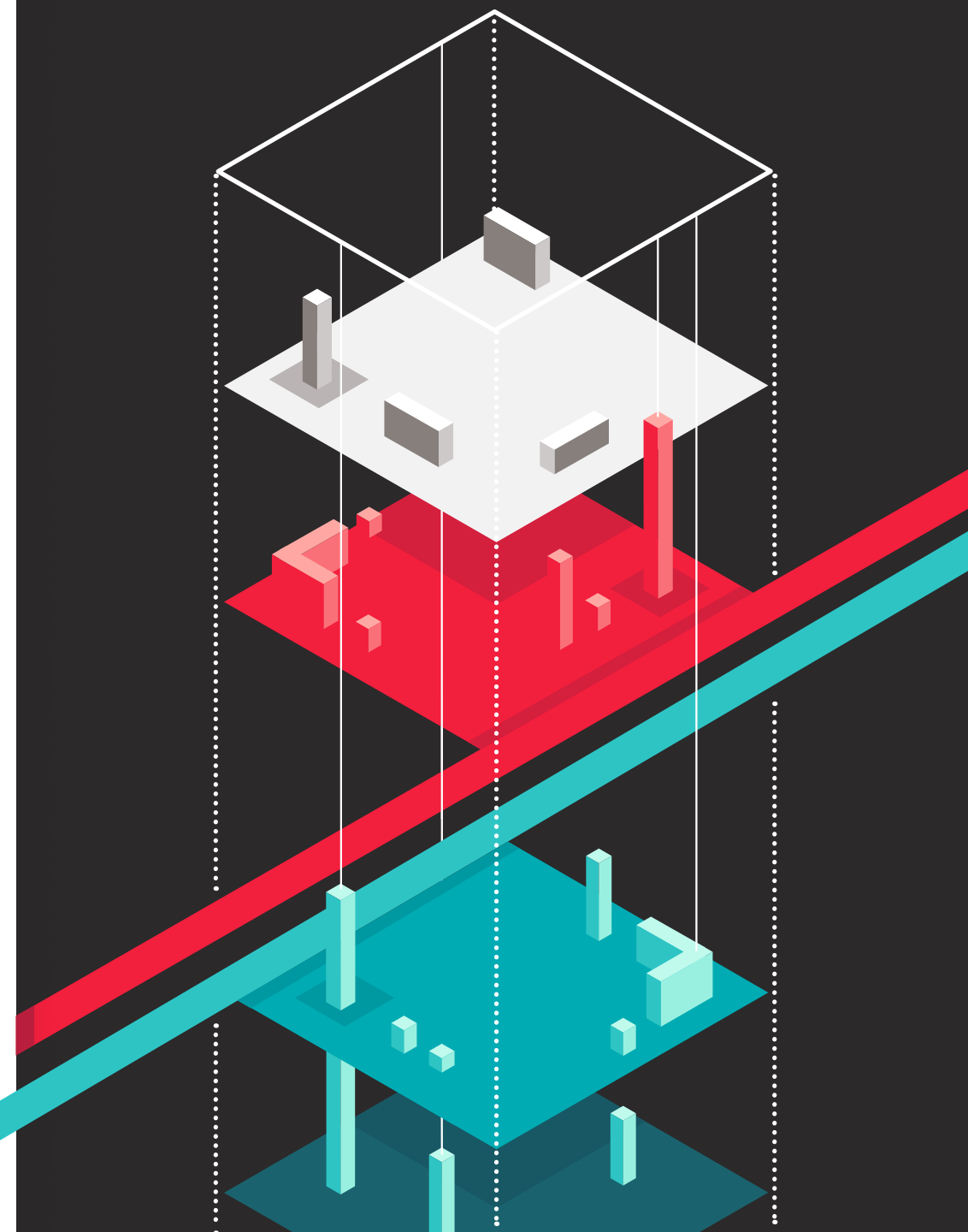
Avoid this peak and valley effect by aligning digital strategy to clear organizational goals.

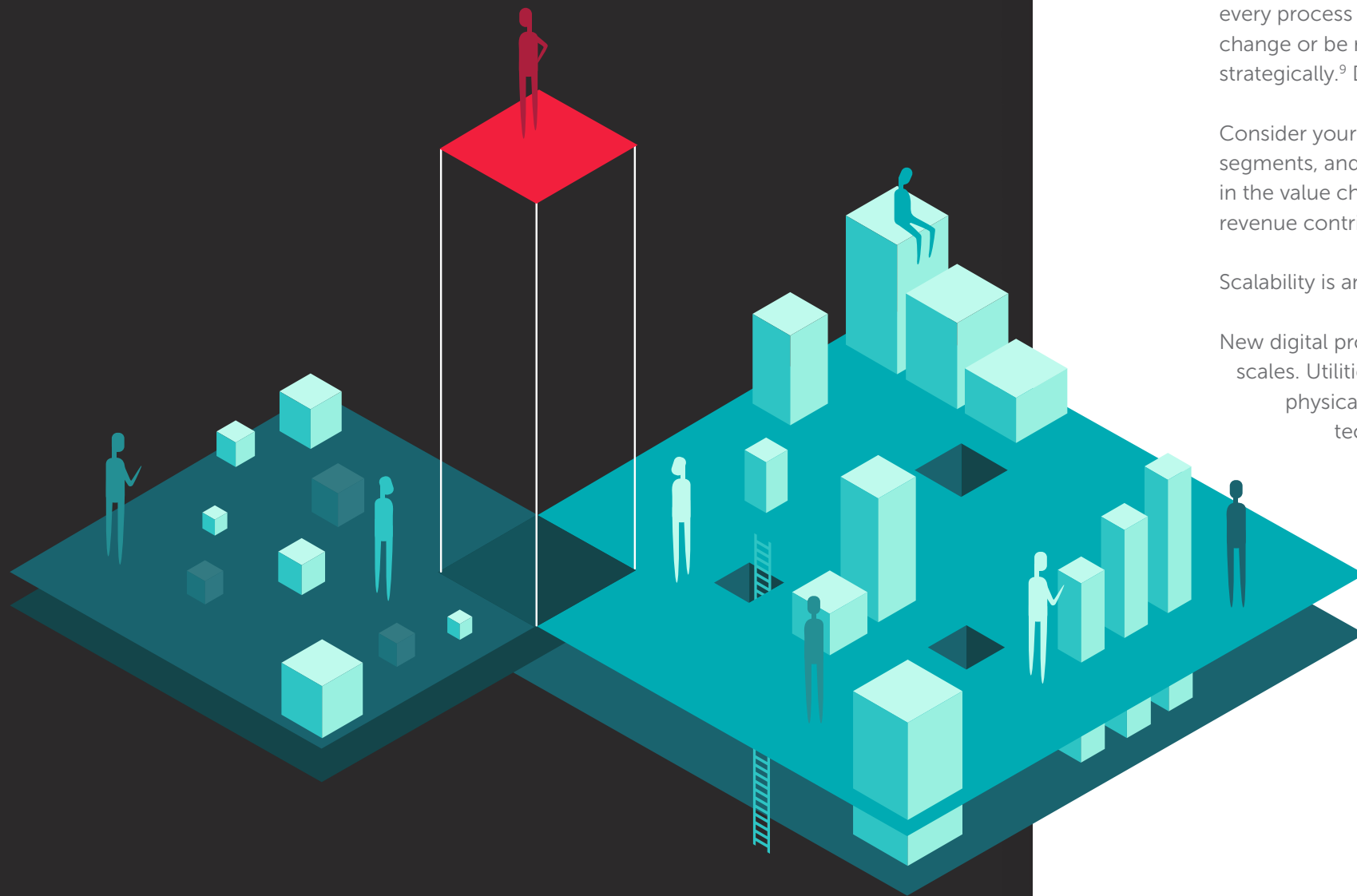
Good objectives include increased revenue, higher profit margins, or improved operational efficiency. Once you've decided on the overarching goal, finesse the particulars until it's both specific and measurable. Here's an example:

Within 6 months of integrating [specific digital strategy], my team will oversee a profit margin increase of [X]%

Making goals measurable is important. If you can't define your progress - good or bad - it's impossible to measure efficacy and easy to lose steam.

Tying digital strategy to business objectives improves your chances of seeing positive change in the long-term.⁸





2

Focus on core processes and scalability

Effective digital transformation doesn't mean rebuilding every process from the ground up. Some processes will change or be replaced, but the key is to make decisions strategically.⁹ Don't try to reinvent the wheel.

Consider your organization carefully, prioritize key segments, and then find the pain points. Focus on, "points in the value chain where digitization would make the largest revenue contribution in the next year or two."¹⁰

Scalability is another important consideration.

New digital processes need to work on both small and large scales. Utilities are big organizations with widely dispersed physical asset networks. To be effective, digital technology and process enhancements must scale up successfully.

3 | Strategize for and commit to the long term

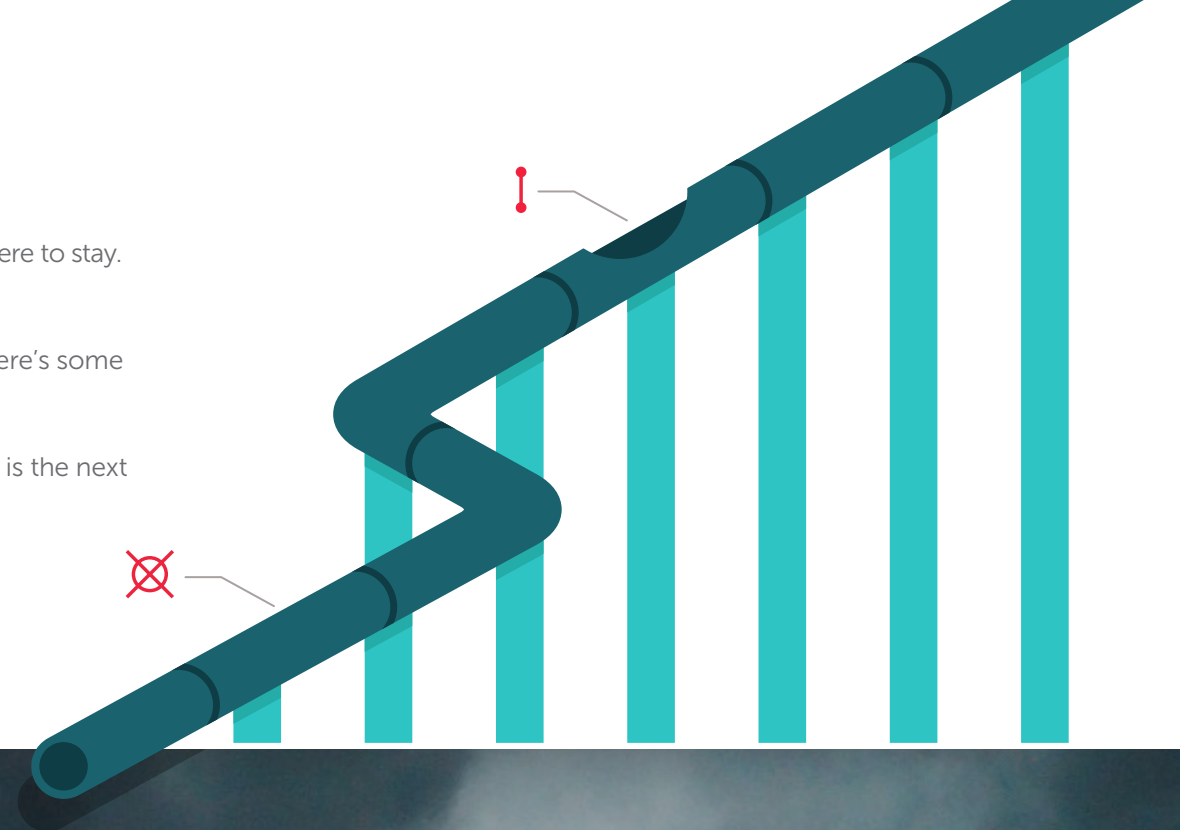
With the sheer quantity of new technologies being developed daily, it should be clear - the digital economy is here to stay. Looking at digitization as a single act is a mistake, one that can have costly consequences in the long term.

Digital transformation should be addressed, encouraged, and integrated as a total cultural shift.¹¹ That said, there's some disagreement in terms of what practical digitization should look like.

Some argue that organizations should aim to be the disruptor. If that's not possible, then being a fast follower is the next best thing. In other words, faster is better.¹² Others recommend a measured, yet consistent approach.¹³

Even with this disparity, there is a common thread: utilities must look to the long term.

Digital transformation is not a destination, it's a state of operations.



Pitfalls to Avoid

One and done mindset

Digital transformation is more than a single act of change, it's a total shift in mindset.

According to Deloitte, "Success [in digital transformation] comes from establishing a clear, repeatable process for digital innovation and change."¹⁴

Effective digital transformation comes from continuous evaluation and re-evaluation of workflows, business operations, and asset management.



Siloed departments

Effective, long-lasting change requires shared goals and open communication between teams.

Making digital transformation the sole responsibility of a single siloed department is a recipe for failure.

It makes it difficult for that department to get traction for their initiatives, and opens the possibility for one change in leadership to derail the entire process.¹⁵



Half measures

To be successful, digital transformation requires resources, commitment, and time. Organizations that are serious about getting ahead of the digital curve must embrace the concept in full.

Half measures lead to expensive, sub-par outcomes. They also give critics the opportunity to point at a failed initiative as a reason why digital transformation shouldn't be a priority.



IV

5 Step Framework for Effective Digital Transformation

1

Build support for digital initiatives at the highest levels¹⁶

Meaningful change must have strong support from leadership. Hiring a C-suite executive tasked with technology and digital initiatives is a good first step. This creates a focal point for any large-scale decisions and signals that digital transformation is a priority.

Next, seek out regular meetings with venture capitalists, entrepreneurs, tech startups, and tech-savvy executives at other utilities. Integrating new technologies into your business is easier when you have a network of individuals who share similar goals, and whose expertise you can tap when necessary.

Finally, spend some time with the board. Encourage them to consider the advantages of digitization, and be sure there's dedicated time in each board meeting for discussion of the organization's ongoing digital strategy.

2

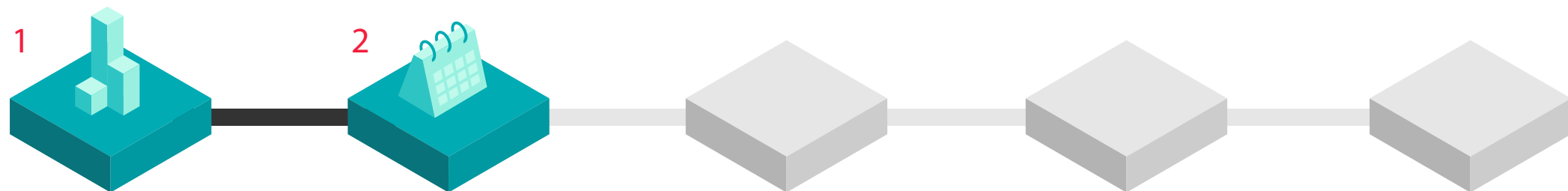
Define both short and long term priorities¹⁷

When starting a new initiative, it's tempting to try addressing everything at once. However, digital transformation is a long process and diving in too fast can lead to burnout and mistakes.

Take a step back and decide your strategy first. Setting priorities - both short and long term - is extremely important.

Choose one aspect of your utility - assets, customer experience, special projects, etc - and focus initial efforts there. Define specific targets with clear, actionable steps, potentially even linking progress to budgets.¹⁸

After deciding your first priority, set a schedule for addressing other items on your wishlist. Of course, priorities may change as time passes, so there's nothing wrong with shifting long term strategy to align with new objectives.



3

Integrate digital workflows and applications¹⁹

With your strategy in mind, start exploring ways to integrate digital workflows and applications. The work done in steps one and two - getting buy-in from senior leadership and deciding priorities - will be important during this stage.

Integrating digital workflows and applications can go one of two ways. The first option is to assemble an in-house team of software designers and developers.²⁰ The second is to outsource to a specialized external vendor.

Which option you choose depends on the kind of applications you need, and the workflows you want to enhance.

The upside of an in-house team is that you control the process from start to finish. The downside is that it's often difficult for these teams to navigate internal bureaucracy and attract top talent.

The advantage of outsourcing to an external vendor is greater specialization and deeper expertise in digital products. External vendors can attract top development talent more easily, and building software is their main focus. Usually this translates to producing a higher quality product faster.

4

Attract and retain digitally fluent employees

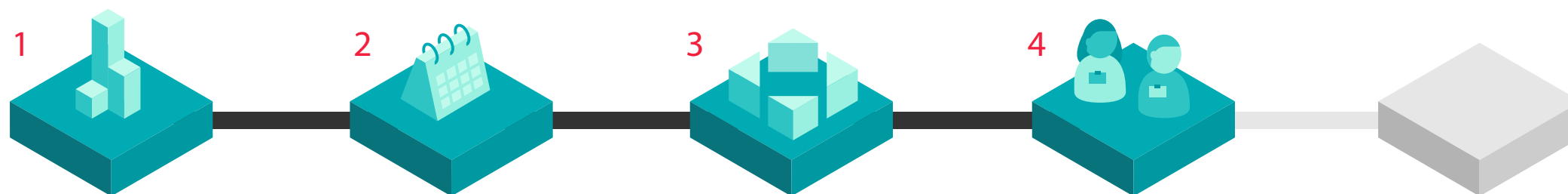
Though support from leadership is key, the day-to-day digital transformation process will be in the hands of employees. This means that attracting the right employees is crucial to success.

Attracting digital talent has, historically, been a challenge for utilities. The main reason is that utilities are often (rightly) thought of as analog organizations, both in terms of their assets and operational processes.

Hiring a CTO is a good start at mitigating this perception issue. It acts as a strong signal that your utility is genuinely interested in moving towards a digital standard.

Another potential strategy is to highlight the intellectual challenges, as well as the social value of facilitating such significant change.²¹

For many young job seekers, millennials especially, having a positive impact on the world is important.²² Framing potential job openings with the idea that efficient utilities benefit communities of every scale could prove fruitful.



5 | Modernize the IT environment

Legacy IT environments are another common challenge. Most utilities have been around for decades, if not longer, and their IT environment reflects that history.

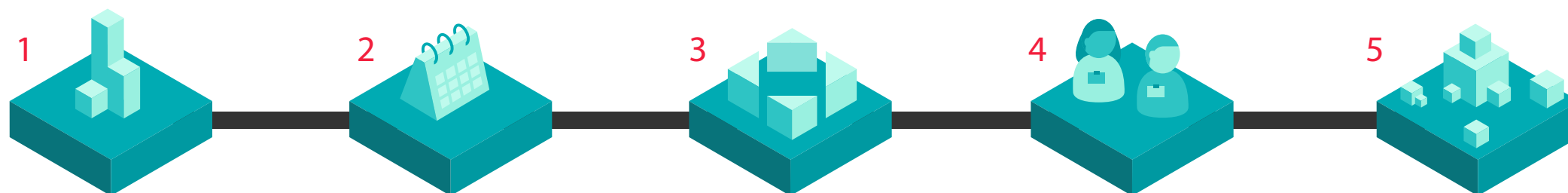
Shifting away from an all-in-one IT approach to a more modular architecture is an important first step. Trying to develop a single platform that addresses every process or workflow is an exercise in frustration and likely failure.

Rather, utilities should use out-of-the-box solutions for tasks such as billing and customer relationships. Then, if a custom application is required for a more complex task, pursue that solution independently.²³

Another important step is to consider your IT processes closely, and take action in one of two ways.

The first option is to streamline and update existing digital processes. Many of these processes may have been conceived in the first or second wave of digital technology, but haven't been upgraded since.

The second option is to digitize core analog processes. This approach is often more difficult, because it involves a heavier airlift in terms of ideation and execution. However, if core processes can be digitized successfully, the results are often positive.²⁴



Securing Quick Wins

We've outlined the golden rules, pitfalls to avoid, and a framework for effective digital transformation. Even with these guidelines, digital transformation can still feel intimidating. You might be asking, where do I start?

Rather than waiting to have all the data, or trying to play out every possible digital transformation scenario, identify opportunities most likely to produce a quick win and start there.

Embrace innovation in the form of "build-assess-learn" cycles.²⁵ Identify a process or pain point, build out a pilot program or MVP prototype, and then evaluate for efficacy.

Don't get wrapped up in a quest for perfection. When taking steps towards digital transformation good is good enough.

IV | A Note on Uneath

Uneath is a modern, map-based project management platform for organizations in the built world. We provide a simpler, more profitable way to manage location-based data: one that's optimized for monitoring and maintaining physical asset networks.

When managed correctly, data enhances your operations and saves millions of dollars. Without proper management, operations slow and revenue is lost.

Uneath delivers on the abandoned promises of legacy GIS software with a map-based project management platform that's simple and runs in the cloud. Use our custom digital toolkits to streamline your daily workflows, increase efficiency, and jumpstart your digital transformation.

Uneath

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To learn more, please reach out in the way that works best for you.

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References

1. "The case for digital reinvention | McKinsey." <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/the-case-for-digital-reinvention>. Accessed 27 Apr. 2020.
2. "Accelerating digital transformations: A playbook for utilities - McKinsey." <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/accelerating-digital-transformations-a-playbook-for-utilities>. Accessed 27 Apr. 2020.
3. "Digital America: A Tale of the Haves and the Have-Mores - McKinsey." 31 Dec. 2015, <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/digital-america-a-tale-of-the-haves-and-have-mores>. Accessed 27 Apr. 2020.
4. "The case for digital reinvention | McKinsey." <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/the-case-for-digital-reinvention>. Accessed 27 Apr. 2020.
5. "Accelerating digital transformations: A playbook for utilities" <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/accelerating-digital-transformations-a-playbook-for-utilities>. Accessed 27 Apr. 2020.
6. "Digital transformation and the utility of the future" 9 Apr. 2019, <https://www2.deloitte.com/us/en/insights/industry/power-and-utilities/digital-transformation-utility-of-the-future.html>. Accessed 27 Apr. 2020.
7. "The digital utility: New opportunities and challenges | McKinsey." <https://www.mckinsey.com/industries/electric-power-and-natural-gas/our-insights/the-digital-utility-new-opportunities-and-challenges>. Accessed 27 Apr. 2020.
8. "The digital utility: New opportunities and challenges | McKinsey." <https://www.mckinsey.com/industries/electric-power-and-natural-gas/our-insights/the-digital-utility-new-opportunities-and-challenges>. Accessed 27 Apr. 2020.
9. "Digital Transformation for Utilities: More Tortoise, Less Hare" 5 Feb. 2019, <https://www.bain.com/insights/digital-transformation-for-utilities-more-tortoise-less-hare/>. Accessed 27 Apr. 2020.
10. "The digital utility: New opportunities and challenges | McKinsey." <https://www.mckinsey.com/industries/electric-power-and-natural-gas/our-insights/the-digital-utility-new-opportunities-and-challenges>. Accessed 27 Apr. 2020.
11. "The digital utility: New opportunities and challenges | McKinsey." <https://www.mckinsey.com/industries/electric-power-and-natural-gas/our-insights/the-digital-utility-new-opportunities-and-challenges>. Accessed 27 Apr. 2020.
12. "The case for digital reinvention | McKinsey." <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/the-case-for-digital-reinvention>. Accessed 27 Apr. 2020.
13. "Digital Transformation for Utilities: More Tortoise, Less Hare" 5 Feb. 2019, <https://www.bain.com/insights/digital-transformation-for-utilities-more-tortoise-less-hare/>. Accessed 27 Apr. 2020.
14. "Digital transformation and the utility of the future | Deloitte" 9 Apr. 2019, <https://www2.deloitte.com/us/en/insights/industry/power-and-utilities/digital-transformation-utility-of-the-future.html>. Accessed 27 Apr. 2020.
15. "Digital transformation and the utility of the future" 9 Apr. 2019, <https://www2.deloitte.com/us/en/insights/industry/power-and-utilities/digital-transformation-utility-of-the-future.html>. Accessed 27 Apr. 2020.
16. "Accelerating digital transformations: A playbook for ... - McKinsey." <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/accelerating-digital-transformations-a-playbook-for-utilities>. Accessed 27 Apr. 2020.
17. "Accelerating digital transformations: A playbook for ... - McKinsey." <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/accelerating-digital-transformations-a-playbook-for-utilities>. Accessed 27 Apr. 2020.
18. "The Digital Utility - McKinsey." 3 Jun. 2018, <https://www.mckinsey.com/~media/McKinsey/Industries/Electric%20Power%20and%20Natural%20Gas/Our%20Insights/The%20Digital%20Utility/The%20Digital%20Utility.ashx>. Accessed 27 Apr. 2020.
19. "Accelerating digital transformations: A playbook for ... - McKinsey." <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/accelerating-digital-transformations-a-playbook-for-utilities>. Accessed 27 Apr. 2020.
20. "Accelerating digital transformations: A playbook for ... - McKinsey." <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/accelerating-digital-transformations-a-playbook-for-utilities>. Accessed 27 Apr. 2020.
21. "Accelerating digital transformations: A playbook for ... - McKinsey." <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/accelerating-digital-transformations-a-playbook-for-utilities>. Accessed 27 Apr. 2020.
22. "Study: Millennials Conflicted in the Pursuit of Meaningful Work." <https://online.olyet.edu/news/study-millennials-conflicted-pursuit-meaningful-work>. Accessed 27 Apr. 2020.
23. "Accelerating digital transformations: A playbook for ... - McKinsey." <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/accelerating-digital-transformations-a-playbook-for-utilities>. Accessed 27 Apr. 2020.
24. "The Digital Utility - McKinsey." 3 Jun. 2018, <https://www.mckinsey.com/~media/McKinsey/Industries/Electric%20Power%20and%20Natural%20Gas/Our%20Insights/The%20Digital%20Utility/The%20Digital%20Utility.ashx>. Accessed 27 Apr. 2020.